# ME 5309 – Mid-Semester Examination

Answer Booklet

**Instructions:** Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

## Question 1 [10 marks]

### Q1(a) [4 marks]

### Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

*Calculation:* $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. $1.2207$ mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

### Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

## Question 2 [10 marks]

### Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\,x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since |sin θ_e| less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to mgℓ = 2.4525 N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \le \sin\theta_e \le 1$, the required torque must lie between $-2.4525$ and $+2.4525$ N m.

### Q2(b) [2 marks]

### Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is Kp(θd-θ), which must balance mgℓ sin θ.

### Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

# Question 4 [10 marks]

## Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

## Q4(b) [2 marks]

With actual b=0.75 and unchanged Nff=8, $k_1$=2, the steady-state relation is θss = bNff/(c+b$k_1$) = 0.75x8/(6+0.75x2)=6/7.5=0.8. For r=1, percentage error = (1-0.8)x100%=20%.

With the actual gain, the steady-state output is $y_{ss} = b_{actual}N_{ff}r/(c+b_{actual}k_1)$ for the stated setup. Substituting the given numbers gives $y_{ss} = 0.8$ for a unit reference. Therefore $e_{ss} = 1 - 0.8 = 0.2$, i.e. a 20% error.

## Q4(c) [4 marks]

With $\dot{\xi}$=r-θ and u=-$k_1$θ-$k_2\dot{\theta}$+K_Iξ, the homogeneous closed-loop characteristic polynomial is $s^3$+(a+b$k_2$)$s^2$+(c+
Desired polynomial: (s+4)[(s+2)²+4]=(s+4)($s^2$+4s+8)=$s^3$+8$s^2$+24s+32. For a=2,b=1,c=6, matching coefficients gives $k_2$=6, $k_1$=18 and K_I=32.

First multiply $(s+4)(s^2+4s+8) = s^3+8s^2+24s+32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of $s$.

## Q4(d) [2 marks]

Integral action gives zero steady-state error if stable. A practical problem is integrator windup.